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# ADVANCES IN HERBAL TRANSDERMAL DRUG DELIVERY SYSTEMS FOR ANTI-INFLAMMATORY THERAPY: FORMULATION, EVALUATION, AND EMERGING TECHNOLOGIES

**Shreya Sahu - Research Scholar | shreyasahu1668@gmail.com**

**Department of Pharmacy, Kalinga University, Raipur, Chhattisgarh**

**Deepesh Kumar - Assistant Professor | deepesh.kumar@kalingauniversity.ac.in**

**Faculty of Pharmacy, Kalinga University, Raipur, Chhattisgarh**

**ABSTRACT:**

*Inflammation is a protective biological response that may progress to chronic disorders if uncontrolled. Conventional therapies such as NSAIDs and corticosteroids are effective but associated with adverse effects during long-term use, creating a need for safer alternatives. Herbal drugs offer multi-targeted anti-inflammatory activity with improved safety; however, their clinical use is limited by poor oral bioavailability and first-pass metabolism. Transdermal drug delivery systems provide an effective alternative by enabling controlled drug release and improved bioavailability. Herbal transdermal patches integrate plant-derived actives with the advantages of transdermal delivery, making them a promising approach for inflammatory conditions. This review summarizes key herbal actives, formulation strategies, and evaluation parameters, along with recent advances such as nanocarrier systems, microneedles, and stimuli-responsive patches. Despite progress, challenges related to stability, skin compatibility, standardization, and regulation remain. Ongoing research is expected to enhance the development of safe and effective herbal transdermal therapies.*

**Keywords:** *Herbal transdermal patches; Anti-inflammatory activity; TDDS; Curcumin; Boswellic acid; Nanocarriers; Microneedles; Penetration enhancers; Drug release; Skin permeation*

## 1. INTRODUCTION

Inflammation is a fundamental biological response to injury, infection, or irritation [1]. While it serves a protective role, its persistent or chronic form is associated with disorders such as arthritis, skin diseases, and musculoskeletal pain [2]. The management of chronic inflammation remains a clinical challenge. Conventional anti-inflammatory agents, including non-steroidal anti-inflammatory drugs (NSAIDs) and corticosteroids, are widely prescribed; however, their long-term oral use is often limited by adverse effects such as gastric irritation, renal toxicity, and cardiovascular risks [3]. In recent years, herbal drugs have emerged as potential alternatives for managing inflammatory conditions. Medicinal plants such as *Curcuma longa*, *Boswellia serrata*, *Capsicum annuum*, and *Withania somnifera* have shown notable anti-inflammatory activity in experimental and clinical studies [4]. Their bioactive constituents exert effects through multiple mechanisms, including inhibition of COX enzymes and suppression of pro-inflammatory cytokines such as TNF-α and IL-6 [5]. Despite these advantages, many herbal compounds suffer from poor oral bioavailability due to low aqueous solubility and extensive first-pass metabolism [6]. This limitation has encouraged the exploration of alternative drug delivery routes, among which the transdermal route is particularly promising, as it bypasses gastrointestinal and hepatic metabolism, maintains stable plasma drug levels, and improves patient compliance [7].

Transdermal drug delivery systems (TDDS), especially patch-based systems, provide controlled and sustained drug release through the skin into systemic circulation [8]. Incorporation of herbal actives into such systems combines the therapeutic potential of plant-derived compounds with improved pharmacokinetic performance [9]. Studies have demonstrated that herbal transdermal patches enhance skin permeation, reduce systemic side effects, and improve therapeutic efficacy [4, 10].

This review provides an overview of herbal transdermal patches for anti-inflammatory therapy, covering key herbal drugs, principles of transdermal delivery, formulation approaches, evaluation parameters, recent advances, and future perspectives.

## 2. HERBAL DRUGS USED FOR ANTI-INFLAMMATORY ACTIVITY

Herbal drugs have been extensively used in traditional systems such as Ayurveda, Unani, and Traditional Chinese Medicine for the management of pain and inflammation [11]. With growing scientific evidence, several plant-derived compounds have been recognized as effective anti-inflammatory agents suitable for incorporation into transdermal patch systems [4].

**2.1 Curcumin:** Curcumin, the bioactive constituent of *Curcuma longa* (turmeric), is among the most widely studied herbal anti-inflammatory agents. It inhibits Nuclear Factor-kappa B (NF-κB), a key regulator of inflammatory gene expression, and suppresses COX-2 along with pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6 [5, 12]. However, its clinical use is limited by very low oral bioavailability (<1%) due to rapid metabolism and poor aqueous solubility, making transdermal delivery a suitable alternative [13].

**2.2 Boswellic Acid:** Boswellic acids, derived from *Boswellia serrata* (shallaki), inhibit the 5-lipoxygenase (5-LOX) pathway, thereby reducing leukotriene-mediated inflammation [14]. They have shown clinical benefits in conditions such as osteoarthritis and rheumatoid arthritis [4]. Transdermal administration may reduce gastrointestinal side effects associated with oral therapy.

**2.3 Capsaicin:** Capsaicin, obtained from *Capsicum annuum*, exerts its effect by depleting Substance P from sensory neurons, leading to reduced pain and inflammation [15]. It is widely used in neuropathic pain, osteoarthritis, and post-herpetic neuralgia. Its availability in marketed topical patches (e.g., Qutenza®) supports its suitability for transdermal delivery [3].

**2.4 Menthol and Camphor:** Menthol (*Mentha piperita*) and camphor (*Cinnamomum camphora*) are commonly used as counter-irritants in topical formulations [16]. Menthol additionally acts as a penetration enhancer by increasing the fluidity of stratum corneum lipids, thereby improving drug permeation [7].

**2.5 Arnica:** *Arnica montana* contains sesquiterpene lactones, particularly helenalin, which inhibit NF-κB activation and reduce inflammatory mediator release [17]. It is traditionally used for bruises, sprains, and muscle soreness and is increasingly incorporated into transdermal systems.

**2.6 Withania somnifera (Ashwagandha):** Withanolides from *Withania somnifera* exhibit anti-inflammatory effects through inhibition of COX enzymes and suppression of pro-inflammatory cytokines [18]. Additionally, its adaptogenic and antioxidant properties make it a promising multi-functional candidate for transdermal formulations.

**Table 1: Summary of Major Herbal Drugs Used in Anti-Inflammatory Transdermal Patches**

| Herbal Drug | Plant Source | Active Constituent | Mechanism of Action |
|---|---|---|---|
| Curcumin | *Curcuma longa* | Curcuminoids | NF-κB inhibition, COX-2 suppression |
| Boswellic Acid | *Boswellia serrata* | AKBA | 5-LOX inhibition |
| Capsaicin | *Capsicum annuum* | Capsaicinoids | Substance P depletion |

| | | | |
|---|---|---|---|
| Menthol | *Mentha piperita* | L-Menthol | Counter-irritant, penetration enhancer |
| Arnica | *Arnica montana* | Helenalin | NF-κB inhibition |
| Ashwagandha | *Withania somnifera* | Withanolides | COX inhibition, cytokine suppression |

## 3. BASICS OF TRANSDERMAL DRUG DELIVERY

The skin, the largest organ of the human body (~1.8 m²), functions as a protective barrier between the internal and external environment [19]. From a drug delivery perspective, it provides an accessible route for systemic administration. Transdermal drug delivery systems (TDDS) are designed to deliver drugs across the skin into systemic circulation in a controlled and predictable manner [8].

**3.1 Structure of Skin and Drug Permeation:** The skin consists of three layers: epidermis, dermis, and hypodermis [20]. The outermost stratum corneum acts as the primary barrier due to its structure of keratinized cells embedded in a lipid matrix, restricting the passage of most molecules [7]. Drug permeation occurs via three pathways: transcellular, intercellular, and appendageal routes [21], with the intercellular pathway being predominant. Permeation is influenced by drug properties, skin condition, and formulation factors [8].

**3.2 Ideal Properties of a Drug for Transdermal Delivery:** Only drugs with suitable physicochemical characteristics can effectively permeate the skin barrier [22].

**Table 2: Ideal physicochemical properties of a drug candidate for transdermal delivery**

| Property | Ideal Value |
|---|---|
| Molecular Weight | < 500 Da |
| Log P (octanol/water) | 1–3 |
| Aqueous Solubility | Sufficient to saturate vehicle |
| Melting Point | < 200°C |
| Daily Dose | < 20 mg/day |
| Skin Irritation | Minimal |

Several herbal actives, including curcumin, boswellic acid, and capsaicin, meet many of these criteria, particularly lipophilicity and relatively low molecular weight, supporting their suitability for transdermal systems [6, 13].

**3.3 Advantages of Transdermal Route:** The transdermal route offers key advantages, especially for drugs with poor oral bioavailability [7, 9]:

- Bypasses first-pass metabolism, improving bioavailability
- Provides sustained and controlled drug release
- Reduces gastrointestinal side effects
- Enhances patient compliance
- Suitable for short half-life drugs
- Non-invasive and self-administrable

**3.4 Limitations of Transdermal Route:** Despite its benefits, certain limitations must be considered [20, 22]:

- Stratum corneum limits permeation of hydrophilic and large molecules
- Suitable only for potent, low-dose drugs
- Risk of skin irritation or sensitization
- Lag time in achieving therapeutic levels
- Inter-individual variability in permeability

These limitations have driven the development of various penetration enhancement strategies.

## 4. HERBAL TRANSDERMAL PATCHES — TYPES AND COMPONENTS

Transdermal patches are multi-layered systems designed to deliver a controlled drug dose through the skin over a defined period [8]. When combined with herbal actives, they provide localized or systemic anti-inflammatory effects with improved compliance and reduced adverse effects [9]. A typical patch consists of multiple functional components that collectively determine performance.

**4.1 Types of Transdermal Patches:** Based on design and drug release mechanism, patches are classified as follows [23]:

**4.1.1 Matrix Type Patches:** In matrix systems, the drug is uniformly dispersed within a polymer matrix that controls release by diffusion [8]. These patches are simple, cost-effective, and widely used in herbal formulations, with studies reporting satisfactory performance for drugs such as curcumin, boswellic acid, and capsaicin [13, 24].

**4.1.2 Reservoir Type Patches:** Reservoir systems contain the drug in a separate compartment, with a rate-controlling membrane regulating release [23]. They provide near zero-order kinetics but are more complex and may pose a risk of dose dumping if damaged [22].

**4.1.3 Drug-in-Adhesive Type Patches:** In this design, the drug is incorporated into the adhesive layer, which also acts as the reservoir [21]. These patches are thin, simple, and ensure good skin contact, making them suitable for potent herbal actives.

**4.2 Components of Herbal Transdermal Patches**

**4.2.1 Backing Membrane:** The backing layer protects the patch from external factors and should be flexible, impermeable, and inert [22]. Common materials include polyester, polyethylene, polyurethane, and aluminium laminates.

**4.2.2 Drug Reservoir or Matrix:** This layer contains the herbal drug along with polymers and excipients. Polymer selection significantly influences drug release [9].

**4.2.3 Polymers:** Polymers form the structural framework and control drug release. Hydrophilic polymers (e.g., HPMC, PVP) enable diffusion-based release, while hydrophobic polymers (e.g., ethylcellulose) provide sustained release [24, 25].

**Table 3: Common polymers used in herbal transdermal patches**

| Polymer | Type | Role |
|---|---|---|
| HPMC (K4M, K15M) | Hydrophilic | Matrix former |
| Ethylcellulose | Hydrophobic | Sustained release |
| PVP K30 | Hydrophilic | Binder, solubility enhancer |
| Eudragit RL/RS | Hydrophobic | Rate control |
| Carbopol 934 | Hydrophilic | Gel matrix |
| Polyvinyl Alcohol | Hydrophilic | Film former |

**4.2.4 Plasticizers:** Plasticizers improve flexibility and strength by lowering the glass transition temperature of polymers [22]. Common examples include PEG 400, propylene glycol, dibutyl phthalate, and triethyl citrate (10–30% w/w) [25].

**4.2.5 Penetration Enhancers:** Penetration enhancers improve drug flux by modifying the stratum corneum barrier [7]. Common agents include oleic acid, propylene glycol, DMSO, tween 80, and terpenes such as menthol and eucalyptol [16].

**4.2.6 Release Liner:** The release liner protects the drug-containing surface and is removed before application. It is typically made of siliconized polyester or fluoropolymer-coated materials [23].

## 5. FORMULATION STRATEGIES

Formulation of herbal transdermal patches requires appropriate selection of polymers, plasticizers, penetration enhancers, and processing methods to achieve optimal drug release and skin permeation [8]. Since many herbal actives exhibit poor solubility and limited permeability, specialized strategies are necessary to enhance their transdermal delivery and achieve therapeutic levels [6, 13].

**5.1 Selection of Polymers and Plasticizers:** Polymer selection plays a key role in determining mechanical properties, drug release, and stability [24]. Hydrophilic polymers such as HPMC (K4M, K15M) are widely used due to their film-forming ability; however, they often produce rapid drug release. Therefore, they are commonly combined with hydrophobic polymers like ethylcellulose to achieve sustained release [25]. Binary HPMC–ethylcellulose systems are extensively studied, where increasing ethylcellulose proportion reduces drug release due to lower water uptake [9, 26]. Plasticizers such as propylene glycol and PEG 400 improve flexibility and also enhance skin hydration, facilitating drug permeation [22].

**5.2 Penetration Enhancement Strategies:** Enhancing drug permeation across the stratum corneum is essential and is achieved using chemical or physical methods [7].

**5.2.1 Chemical Penetration Enhancers:** Chemical enhancers temporarily alter lipid organization in the stratum corneum, increasing permeability [21]. Common examples include:

- Oleic acid – disrupts lipid bilayers; effective for lipophilic drugs [13]
- Propylene glycol – improves solubility and acts synergistically [22]
- Tween 80 – enhances drug partitioning
- Menthol and eucalyptol – natural terpenes that increase lipid fluidity [16]
- DMSO – highly effective but used cautiously due to irritation [7]

**5.2.2 Physical Penetration Enhancement:** Physical techniques enhance permeation without altering formulation composition [28]:

- Microneedles – create micropores for drug entry [29]
- Iontophoresis – uses electrical current to drive molecules
- Sonophoresis – ultrasound disrupts skin barrier [7]

**5.3 Nanocarrier-Based Herbal Patch Systems:** Nanocarriers improve solubility, stability, and skin permeation of herbal actives [30].

- **Nanoemulsions:** Enhance permeation by increasing drug availability and reducing interfacial resistance [13, 30]
- **Liposomes/Transfersomes:** Vesicular systems; transfersomes provide superior deformability and penetration [32]
- **Niosomes:** Stable and cost-effective vesicles offering sustained release [33]
- **Solid Lipid Nanoparticles (SLNs):** Improve stability and provide controlled release [30, 34]

**5.4 Solvent Casting Method:** Solvent casting is the most commonly used laboratory method for patch preparation [25]. Polymers are dissolved in a suitable solvent, followed by addition of drug and excipients, then cast and dried to form films. Patch quality depends on variables such as solvent type, drying conditions, and casting thickness [22, 25].

## 6. EVALUATION PARAMETERS

Evaluation ensures quality, performance, and reproducibility of herbal patches [22], and includes physicochemical and biological assessments.

### 6.1 Physicochemical Evaluation

- **Thickness:** Measured using a micrometer to ensure uniform drug distribution (typically 0.2–0.5 mm) [24, 25]
- **Weight Uniformity:** Variation within ±5% indicates uniform composition [22]
- **Folding Endurance:** Values >300 indicate good flexibility [25]
- **Moisture Content/Uptake:** Low moisture (<3%) preferred for stability, though slight moisture prevents brittleness [22, 25]
- **Mechanical Properties:** Tensile strength and elongation assess durability [26]
- **Drug Content:** 95–105% indicates acceptable uniformity; HPLC preferred for herbal systems [13, 24]

**Table 4: Physicochemical Evaluation Parameters**

| Parameter | Criteria | Method |
|---|---|---|
| Thickness | Uniform | Micrometer |
| Weight | ±5% | Balance |
| Folding Endurance | >300 | Manual |
| Moisture Content | <3% | Oven |
| Moisture Uptake | Recorded | Desiccator |
| Tensile Strength | Adequate | Texture analyzer |
| Drug Content | 95–105% | UV/HPLC |
| Surface pH | 5.5–7.0 | pH meter |

**6.2 In Vitro Drug Release:** Drug release is commonly evaluated using the Franz diffusion cell, where the patch is placed on a membrane separating donor and receptor compartments [35]. Samples are collected over time and analyzed to determine release kinetics [24, 35].

**Table 5: Kinetic Models for Drug Release**

| Model | Equation | Mechanism |
|---|---|---|
| Zero-order | $Qt = Q0 + K0t$ | Constant release |
| First-order | $\ln Qt = \ln Q0 - K1t$ | Concentration-dependent |
| Higuchi | $Qt = KH\sqrt{t}$ | Diffusion |
| Korsmeyer-Peppas | $Mt/M\infty = Kt^n$ | Diffusion/erosion |

**6.3 Ex Vivo Skin Permeation:** Ex vivo studies using animal or human skin provide realistic permeation data, including flux and permeability coefficient [7, 36].

**6.4 Skin Irritation Study:** Skin compatibility is assessed by observing erythema and edema, and calculating irritation index [37].

**6.5 In Vivo Anti-Inflammatory Activity:** Efficacy is commonly evaluated using the carrageenan-induced paw edema model. Reduction in edema confirms anti-inflammatory activity of herbal patches [13, 14, 38].

## 7. RECENT ADVANCES IN HERBAL TRANSDERMAL PATCHES FOR ANTI-INFLAMMATORY THERAPY

Herbal transdermal drug delivery has advanced significantly, moving beyond conventional systems toward technologies that enhance permeation, precision, and therapeutic efficiency [28].

**7.1 Microneedle-Based Herbal Patches:** Microneedles (25–900 μm) are integrated into patches to bypass the stratum corneum by creating temporary microchannels, enabling delivery of poorly permeable molecules without pain [28, 29]. Types include:

- Solid microneedles – create pores before drug application
- Dissolving microneedles – release drug upon polymer dissolution
- Hollow microneedles – allow controlled drug infusion

Curcumin-loaded microneedle patches have shown improved permeation and enhanced anti-inflammatory activity compared to conventional systems [29, 39].

**7.2 Nano-Herbal Combination Patches:** These systems incorporate nanocarrier-loaded herbal actives into patch matrices, improving stability, permeation, and controlled release [30, 34]. Enhanced performance has been reported for curcumin transfersome and boswellic acid niosome patches compared to conventional formulations [17, 32, 33].

**7.3 3D Printed Transdermal Patches:** 3D printing enables fabrication of patches with controlled geometry, drug loading, and release characteristics [40]. Techniques such as fused deposition modeling and inkjet printing allow precise and customizable designs. Herbal actives like curcumin and menthol have demonstrated uniform distribution and controlled release in 3D-printed systems [40, 41].

**7.4 Stimuli-Responsive Smart Patches:** Smart patches release drugs in response to specific triggers, improving site-specific delivery [42]:

- pH-responsive systems for inflamed tissue
- Thermo-responsive polymers activated at body temperature
- ROS-responsive systems targeting oxidative stress

These systems enhance therapeutic efficiency by synchronizing drug release with physiological conditions.

**7.5 Ethosomes and Cubosomes:** Ethosomes, with high ethanol content, improve deformability and skin penetration, enhancing drug delivery [44, 45]. Cubosomes offer high drug-loading capacity and sustained release, with promising results reported for boswellic acid formulations [45].

**7.6 Co-delivery of Multiple Herbal Actives:** Co-delivery strategies improve therapeutic outcomes through synergistic effects [4]. Examples include:

- Curcumin + boswellic acid → dual pathway inhibition
- Menthol + herbal actives → enhanced permeation

This approach aligns with polyherbal therapeutic concepts [11].

## 8. CHALLENGES AND FUTURE PERSPECTIVES

Despite progress, several challenges remain. Skin irritation may occur due to certain herbal extracts or enhancers during prolonged application. Stability is another concern, as many herbal compounds are sensitive to environmental factors such as light, heat, and oxygen, requiring appropriate formulation and packaging strategies [40]. Variability in herbal composition due to differences in plant source and processing affects reproducibility and efficacy, highlighting the need for proper standardization and quality control [44]. Regulatory limitations also exist, as clear guidelines for herbal transdermal systems are still evolving, complicating approval processes [43].

Scale-up from laboratory to industrial production presents additional challenges in maintaining consistency and performance [45]. However, advances in nanocarriers, penetration enhancement techniques, and formulation strategies are expected to address these issues. Improved standardization and regulatory clarity will further support the development of herbal transdermal systems [45].

## 9. CONCLUSION

Herbal transdermal patches offer a promising approach for managing inflammatory conditions by combining plant-based therapeutics with the advantages of transdermal delivery, including controlled release, improved compliance, and avoidance of first-pass metabolism. Herbal actives such as curcumin and boswellic acid have demonstrated effective anti-inflammatory activity in these systems. Successful formulation depends on appropriate selection of polymers, enhancers, and processing techniques, along with thorough evaluation of physicochemical and biological properties. Recent advances, including nanocarrier-based delivery, microneedle systems, and stimuli-responsive patches, have further strengthened their potential. Although

challenges related to stability, standardization, and regulation persist, ongoing research is expected to establish herbal transdermal patches as an effective and widely accepted strategy for anti-inflammatory therapy.

## 10. REFERENCES

1. Jayakumar, R.; Prabaharan, M.; Sudheesh Kumar, P.T.; Nair, S.V.; Tamura, H. Biomaterials Based on Chitin and Chitosan in Wound Dressing Applications. *Biotechnol. Adv.* 2011, 29, 322–337.
2. Marks, R. The Stratum Corneum Barrier: The Final Frontier. *J. Nutr.* 2004, 134, 2017S–2021S.
3. Elias, P.M. Epidermal Lipids, Barrier Function, and Desquamation. *J. Investig. Dermatol.* 1983, 80, S44–S49.
4. Scheuplein, R.J.; Blank, I.H. Permeability of the Skin. *Physiol. Rev.* 1971, 51, 702–747.
5. Pouillot, A.; Dayan, N.; Polla, A.S.; Polla, L.L.; Polla, B.S. The Stratum Corneum: A Double Paradox. *J. Cosmet. Dermatol.* 2008, 7, 143–148.
6. Barry, B.W. Lipid-Protein-Partitioning Theory of Skin Penetration Enhancement. *J. Control. Release* 1991, 15, 237–248.
7. Barry, B.W. Novel Mechanisms and Devices to Enable Successful Transdermal Drug Delivery. *Eur. J. Pharm. Sci.* 2001, 14, 101–114.
8. Wokovich, A.M.; Prodduturi, S.; Doub, W.H.; Hussain, A.S.; Buhse, L.F. Transdermal Drug Delivery System Adhesion as a Critical Attribute. *Eur. J. Pharm. Biopharm.* 2006, 64, 1–8.
9. Prausnitz, M.R.; Langer, R. Transdermal Drug Delivery. *Nat. Biotechnol.* 2008, 26, 1261–1268.
10. Alexander, A.; Dwivedi, S.; Ajazuddin; Giri, T.K.; Saraf, S.; Saraf, S.; Tripathi, D.K. Approaches for Enhancing Transdermal Drug Delivery. *J. Control. Release* 2012, 164, 26–40.
11. Nastiti, C.; Ponto, T.; Abd, E.; Grice, J.; Benson, H.; Roberts, M. Topical Nano and Microemulsions for Skin Delivery. *Pharmaceutics* 2017, 9, 37.
12. Roberts, M.; Mohammed, Y.; Pastore, M.; et al. Topical and Cutaneous Delivery Using Nanosystems. *J. Control. Release* 2017, 247, 86–105.
13. Caló, E.; Khutoryanskiy, V.V. Biomedical Applications of Hydrogels. *Eur. Polym. J.* 2015, 65, 252–267.
14. Ahsan, A.; Tian, W.-X.; Farooq, M.A.; Khan, D.H. Hydrogels in Transdermal Drug Delivery. *Int. J. Polym. Mater.* 2021, 70, 574–584.
15. Barrientos, S.; Stojadinovic, O.; Golinko, M.S.; Brem, H.; Tomic-Canic, M. Growth Factors in Wound Healing. *Wound Repair Regen.* 2008, 16, 585–601.
16. Nathan, C. Points of Control in Inflammation. *Nature* 2002, 420, 846–852.

**17.** Gurtner, G.C.; Werner, S.; Barrandon, Y.; Longaker, M.T. Wound Repair and Regeneration. *Nature* 2008, 453, 314–321.

**18.** Anderson, K.; Hamm, R.L. Factors That Impair Wound Healing. *J. Am. Coll. Clin. Wound Spec.* 2012, 4, 84–91.

**19.** Elias, P.M.; Wood, L.C.; Feingold, K.R. Epidermal Pathogenesis of Dermatoses. *Am. J. Contact Dermat.* 1999, 10, 119–126.

**20.** Fukumoto, A.; Tajima, K.; Hori, M.; et al. Analgesic Effect of Flurbiprofen Plaster. *J. Pharm. Pharmacol.* 2018, 70, 929–936.

**21.** Irvine, J.; Afrose, A.; Islam, N. Ibuprofen Delivery Strategies. *Drug Dev. Ind. Pharm.* 2018, 44, 173–183.

**22.** DuBois, R.N.; Abramson, S.B.; Crofford, L.; et al. Cyclooxygenase in Biology. *FASEB J.* 1998, 12, 1063–1073.

**23.** Kuczyńska, J.; Nieradko-Iwanicka, B. Ketoprofen Safety. *Biomed. Pharmacother.* 2021, 139, 111608.

**24.** Warner, T.D.; Giuliano, F.; Vojnovic, I.; et al. NSAIDs and GI Toxicity. *Proc. Natl. Acad. Sci. USA* 1999, 96, 7563–7568.

**25.** Yuan, C.; Sidhu, R.S.; Kuklev, D.V.; et al. Cyclooxygenase Allosterism. *J. Biol. Chem.* 2009, 284, 10046–10055.

**26.** Ricciotti, E.; FitzGerald, G.A. Prostaglandins and Inflammation. *ATVB* 2011, 31, 986–1000.

**27.** Meek, I.L.; Van de Laar, M.A.; Vonkeman, H.E. NSAIDs and Cardiovascular Risk. *Pharmaceuticals* 2010, 3, 2146–2162.

**28.** Sánchez-Borges, M.; Capriles-Hulett, A.; Caballero-Fonseca, F. Skin Reactions to Ibuprofen. *Expert Opin. Drug Saf.* 2005, 4, 837–848.

**29.** Noize, P.; Bénard-Laribière, A.; et al. Ketoprofen Cutaneous Effects. *Am. J. Clin. Dermatol.* 2010, 11, 131–136.

**30.** Busti, A.J.; Hooper, J.S.; Amaya, C.J.; Kazi, S. Anti-inflammatory Therapy and Wound Healing. *Pharmacotherapy* 2005, 25, 1566–1591.

**31.** Ulbrich, H.; Dannhardt, G. NSAID Classification. *Pharm. Unserer Zeit* 2002, 31, 146–154.

**32.** Patel, A.; Bell, M.; O'Connor, C.; et al. Ibuprofen Skin Delivery. *Int. J. Pharm.* 2013, 457, 9–13.

**33.** Seth, P.L. Percutaneous Absorption of Ibuprofen. *Arzneimittelforschung* 1993, 43, 919–921.

**34.** Berner, G.; Engels, B.; Vögtle-Junkert, U. Ibuprofen Therapy Studies. *Drugs Exp. Clin. Res.* 1989, 15, 559–564.

**35.** Tegeder, I.; Muth-Selbach, U.; Lötsch, J.; et al. Tissue Concentration Studies of Ibuprofen. *Clin. Pharmacol. Ther.* 1999, 65, 357–368.

**36.** Brown, B.R.; Carter, W.C. *Ibuprofen: Clinical Pharmacology*; Nova Science Publishers: New York, 2013.

**37.** Hadgraft, J.; Whitefield, M.; Rosher, P.H. Skin Penetration of Ibuprofen. *Skin Pharmacol. Physiol.* 2003, 16, 137–142.

**38.** Bagheri, H.; Lhiaubet, V.; Montastruc, J.L.; Chouini-Lalanne, N. Photosensitivity to Ketoprofen. *Drug Saf.* 2000, 22, 339–349.

**39.** Carbone, C.; Rende, P.; et al. Ketoprofen Safety. *J. Pharmacol. Pharmacother.* 2013, 4, S99–S103.

**40.** Coaccioli, S. Ketoprofen Gel Overview. *Eur. Rev. Med. Pharmacol. Sci.* 2011, 15, 943–949.

**41.** Rafanan, B.S.; Valdecañas, B.F.; et al. Osteoarthritis Pain Management. *Pain Manag.* 2018, 8, 115–128.

**42.** Zhao, M.; Yao, J.; Meng, X.; et al. Nanoparticles for Rheumatoid Arthritis. *J. Pharm. Sci.* 2021, 110, 2233–2240.

**43.** Atzeni, F.; Masala, I.F.; et al. Ketoprofen vs Ibuprofen. *Pain Ther.* 2021, 10, 577–588.

**44.** Wang, J.; Zhao, S.; Zhang, M.; He, B. Eco-Pharmacovigilance of Ketoprofen. *Chemosphere* 2018, 194, 450–462.

**45.** Veys, E.M. Experience with Ketoprofen. *Scand. J. Rheumatol. Suppl.* 1991, 90, 3–44.